

5 Biodiversity can be considered at three different levels.

Fig. 5.1, Fig. 5.2 and Fig. 5.3 are images that show these three different levels of biodiversity.



Fig. 5.1



Fig. 5.2



Fig. 5.3

- (a) Describe the level of biodiversity most clearly visible in each image.

Fig. 5.1

Fig. 5.2

Fig. 5.3

[3]

- (b) Prairie strips are restored habitat areas planted with native grasses and wildflowers. Prairie strips are usually located between or around the edges of fields of maize or soy bean.

Students in North America wanted to investigate biodiversity along a prairie strip.

- (i) Outline how the students could measure the biodiversity of plants and insects along a prairie strip.

plants

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insects

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[4]

- (ii) Mathematical methods and statistical tests help to summarise and compare biodiversity data.

Name the mathematical method **or** statistical test that the students can use:

to estimate the total biodiversity of a prairie strip

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to test whether plant biodiversity is associated with insect biodiversity.

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[2]

(c) Prairie strips have environmental and economic effects.

Scientists have made long-term measurements of the environmental effects of prairie strips.

The scientists found that when prairie strips formed 10% of a crop-field:

- soil erosion decreased by 95%
- mineral loss from the field decreased by 90% for phosphorus compounds
- mineral loss from the field decreased by 85% for nitrogen compounds.

Crop plants need phosphorus (P) and nitrogen (N) to grow.

Evaluate the **economic** effects of prairie strips on farming.

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[Total: 12]

Question 6 starts on page 18.